



Indefinite Integration

But just as much as it is easy to find the differential of a given quantity, so it is difficult to find the integral of a given differential. Moreover, sometimes we cannot say with certainty whether the integral of a given quantity can be found or not. Bernoulli, Johann

If f & g are functions of x such that $g'(x) = f(x)$, then indefinite integration of $f(x)$ with respect to x is defined and denoted as $\int f(x) dx = g(x) + C$, where C is called the **constant of integration**.

Standard Formula:

$$(i) \quad \int (ax + b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C, n \neq -1$$

$$(ii) \quad \int \frac{dx}{ax+b} = \frac{1}{a} \ln |ax+b| + C$$

$$(iii) \quad \int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$(iv) \quad \int a^{px+q} dx = \frac{1}{p} \frac{a^{px+q}}{\ln a} + C; a > 0$$

$$(v) \quad \int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$$

$$(vi) \quad \int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$$

$$(vii) \quad \int \tan(ax+b) dx = \frac{1}{a} \ln |\sec(ax+b)| + C$$

$$(viii) \quad \int \cot(ax+b) dx = \frac{1}{a} \ln |\sin(ax+b)| + C$$

$$(ix) \quad \int \sec^2(ax+b) dx = \frac{1}{a} \tan(ax+b) + C$$

$$(x) \quad \int \operatorname{cosec}^2(ax+b) dx = -\frac{1}{a} \cot(ax+b) + C$$

$$(xi) \quad \int \sec(ax+b) \cdot \tan(ax+b) dx = \frac{1}{a} \sec(ax+b) + C$$

$$(xii) \quad \int \operatorname{cosec}(ax+b) \cdot \cot(ax+b) dx = -\frac{1}{a} \operatorname{cosec}(ax+b) + C$$

$$(xiii) \quad \int \sec x dx = \ln |\sec x + \tan x| + C \quad \text{OR} \quad \ln \left| \tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right| + C$$

$$(xiv) \quad \int \operatorname{cosec} x dx = \ln |\operatorname{cosec} x - \cot x| + C \quad \text{OR} \quad \ln \left| \tan \frac{x}{2} \right| + C \quad \text{OR} \quad -\ln |\operatorname{cosec} x + \cot x| + C$$

$$(xv) \quad \int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a} + C$$





$$(xvi) \quad \int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

$$(xvii) \quad \int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a} \sec^{-1} \frac{x}{a} + C$$

$$(xviii) \quad \int \frac{dx}{\sqrt{x^2 + a^2}} = \ln \left| x + \sqrt{x^2 + a^2} \right| + C \quad \text{OR} \quad \sinh^{-1} \frac{x}{a} + C$$

$$(xix) \quad \int \frac{dx}{\sqrt{x^2 - a^2}} = \ln \left| x + \sqrt{x^2 - a^2} \right| + C \quad \text{OR} \quad \cosh^{-1} \frac{x}{a} + C$$

$$(xx) \quad \int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C$$

$$(xxi) \quad \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C$$

$$(xxii) \quad \int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C$$

$$(xxiii) \quad \int \sqrt{x^2 + a^2} dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \ln \left| \frac{x + \sqrt{x^2 + a^2}}{a} \right| + C$$

$$(xxiv) \quad \int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \ln \left| \frac{x + \sqrt{x^2 - a^2}}{a} \right| + C$$

$$(xxv) \quad \int e^{ax} \cdot \sin bx dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) + C$$

$$(xxvi) \quad \int e^{ax} \cdot \cos bx dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) + C$$

Theorems on integration

$$(i) \quad \int C f(x) \cdot dx = C \int f(x) \cdot dx$$

$$(ii) \quad \int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$$

$$(iii) \quad \int f(x) dx = g(x) + C_1 \Rightarrow \int f(ax+b) dx = \frac{g(ax+b)}{a} + C_2$$

Example # 1 Evaluate : $\int 3x^6 dx$

Solution : $\int 3x^6 dx = \frac{3}{7} x^7 + C$





Example # 2 Evaluate : $\int \left(x^3 + 5x^2 - 4 + \frac{7}{x} + \frac{2}{\sqrt{x}} \right) dx$

Solution :

$$\begin{aligned} & \int \left(x^3 + 5x^2 - 4 + \frac{7}{x} + \frac{2}{\sqrt{x}} \right) dx \\ &= \int x^3 dx + \int 5x^2 dx - \int 4 dx + \int \frac{7}{x} dx + \int \frac{2}{\sqrt{x}} dx \\ &= \int x^3 dx + 5 \int x^2 dx - 4 \int 1 dx + 7 \int \frac{1}{x} dx + 2 \int x^{-1/2} dx \\ &= \frac{x^4}{4} + 5 \cdot \frac{x^3}{3} - 4x + 7 \ln |x| + 2 \left(\frac{x^{1/2}}{1/2} \right) + C = \frac{x^4}{4} + \frac{5}{3} x^3 - 4x + 7 \ln |x| + 4\sqrt{x} + C \end{aligned}$$

Example # 3 Evaluate : $\int 2^{x \log_2 3} dx$

Solution : We have, $\int 2^{x \log_2 3} dx = \int 3^x dx = \frac{3^x}{\ln 3} + C$

Example # 4 Evaluate : $\int \frac{4^x + 5^x}{7^x} dx$

Solution :

$$\int \frac{4^x + 5^x}{7^x} dx = \int \left(\frac{4^x}{7^x} + \frac{5^x}{7^x} \right) dx = \int \left[\left(\frac{4}{7} \right)^x + \left(\frac{5}{7} \right)^x \right] dx = \frac{(4/7)^x}{\ln \left(\frac{4}{7} \right)} + \frac{(5/7)^x}{\ln \left(\frac{5}{7} \right)} + C$$

Example # 5 Evaluate : $\int \frac{\cos 7x - \cos 8x}{1 + 2 \cos 5x} dx$

Solution : We have,

$$\begin{aligned} \int \frac{\cos 7x - \cos 8x}{1 + 2 \cos 5x} dx &= \frac{1}{2} \int \frac{2 \sin \frac{5x}{2} \cos 7x - 2 \sin \frac{5x}{2} \cos 8x}{\sin \frac{5x}{2} + 2 \cos 5x \sin \frac{5x}{2}} dx \\ &= \frac{1}{2} \int \frac{\left(\sin \frac{19x}{2} - \sin \frac{9x}{2} \right) - \left(\sin \frac{21x}{2} - \sin \frac{11x}{2} \right)}{\sin \frac{5x}{2} + \sin \frac{15x}{2} - \sin \frac{5x}{2}} dx \\ &= \frac{1}{2} \int \frac{\left(\sin \frac{19x}{2} + \sin \frac{11x}{2} \right) - \left(\sin \frac{21x}{2} + \sin \frac{9x}{2} \right)}{\sin \frac{15x}{2}} dx \\ &= \frac{1}{2} \int \frac{2 \sin \frac{15x}{2} \cos 2x - 2 \sin \frac{15x}{2} \cos 3x}{\sin \frac{15x}{2}} dx = \int (\cos 2x - \cos 3x) dx = \frac{1}{2} \sin 2x - \frac{1}{3} \sin 3x + C \end{aligned}$$

Example # 6 Evaluate : $\int \frac{x^3}{(x+1)^2} dx$

Solution :

$$\begin{aligned} \int \frac{x^3}{(x+1)^2} dx &= \int \frac{x^3 + 1 - 1}{(x+1)^2} dx = \int \frac{x^3 + 1}{(x+1)^2} dx - \int \frac{1}{(x+1)^2} dx \\ &= \int \frac{(x+1)(x^2 - x + 1)}{(x+1)^2} dx - \int \frac{1}{(x+1)^2} dx = \int \frac{x^2 - x + 1}{(x+1)} dx - \int \frac{1}{(x+1)^2} dx \\ &= \int \left(x - 2 + \frac{3}{x+1} \right) dx - \int \frac{1}{(x+1)^2} dx = \frac{x^2}{2} - 2x + 3 \ln(x+1) + \frac{1}{x+1} + C \end{aligned}$$





Example # 7 : Evaluate : $\int \frac{1}{4+9x^2} dx$

Solution : We have

$$\begin{aligned}\int \frac{1}{4+9x^2} dx &= dx \cdot \frac{1}{9} \int \frac{1}{\frac{4}{9}+x^2} = \frac{1}{9} \int \frac{1}{(2/3)^2+x^2} dx \\ &= \frac{1}{9} \cdot \frac{1}{(2/3)} \tan^{-1} \left(\frac{x}{2/3} \right) + C = \frac{1}{6} \tan^{-1} \left(\frac{3x}{2} \right) + C\end{aligned}$$

Example # 8 : Evaluate : $\int \cos x \cos 2x dx$

Solution : $\int \cos x \cos 2x dx = \frac{1}{2} \int 2 \cos x \cos 2x dx = \frac{1}{2} \int (\cos 3x + \cos x) dx = \frac{1}{2} \left(\frac{\sin 3x}{3} + \sin x \right) + C$

Self Practice Problems :

(1) Evaluate : $\int \tan^2 x dx$

(2) Evaluate : $\int \frac{1}{1+\sin x} dx$

Ans. (1) $\tan x - x + C$ (2) $\tan x - \sec x + C$

Integration by Substitution

If we substitute $\phi(x) = t$ in an integral then

- (i) everywhere x will be replaced in terms of new variable t .
- (ii) dx also gets converted in terms of dt .

Example # 9 : Evaluate : $\int \frac{\cos x + x \sin x}{x(x + \cos x)} dx$

Solution : We have,

$$\begin{aligned}&\int \frac{\cos x + x \sin x}{x(x + \cos x)} dx \\ &= \int \frac{(x + \cos x) - x + x \sin x}{x(x + \cos x)} dx = \int \left(\frac{1}{x} - \frac{1 - \sin x}{x + \cos x} \right) dx = \int \frac{1}{x} dx - \int \frac{1 - \sin x}{x + \cos x} dx \\ &= \int \frac{1}{x} dx - \int \frac{1}{x + \cos x} d(x + \cos x) = \ln|x| - \ln|x + \cos x| + C.\end{aligned}$$

Example # 10 : Evaluate : $\int \frac{(\ln x)^n}{x} dx$

Solution : We have $\int \frac{(\ln x)^n}{x} dx = \int (\ln x)^n \frac{1}{x} dx = \int (\ln x)^n d(\ln x) = \frac{(\ln x)^{n+1}}{n+1} + C$

Example # 11 : Evaluate : $\int \frac{(\sin^{-1} x)^3}{\sqrt{1-x^2}} dx$

Solution : We have, $\int \frac{(\sin^{-1} x)^3}{\sqrt{1-x^2}} dx = \int (\sin^{-1} x)^3 d(\sin^{-1} x) = \frac{(\sin^{-1} x)^4}{4} + C$





Example # 12 : Evaluate : $\int \frac{x}{x^4 + 2x^2 + 2} dx$

Solution : We have,

$$I = \int \frac{x}{x^4 + 2x^2 + 2} dx = \int \frac{x}{(x^2)^2 + 2x^2 + 2} dx \quad \left\{ \text{Put } x^2 = t \Rightarrow x \cdot dx = \frac{dt}{2} \right\}$$

$$\Rightarrow I = \frac{1}{2} \int \frac{1}{t^2 + 2t + 2} dt = \frac{1}{2} \int \frac{1}{(t+1)^2 + 1} dt = \frac{1}{2} \tan^{-1}(t+1) + C$$

$$= \frac{1}{2} \tan^{-1}(x^2 + 1) + C$$

Note: (i) $\int [f(x)]^n f'(x) dx = \frac{(f(x))^{n+1}}{n+1} + C$

(ii) $\int \frac{f'(x)}{[f(x)]^n} dx = \frac{(f(x))^{1-n}}{1-n} + C, n \neq 1$

(iii) $\int \frac{dx}{x(x^n+1)}$; $n \in \mathbb{N}$ Take x^n common & put $1+x^{-n}=t$.

(iv) $\int \frac{dx}{x^2(x^n+1)^{(n-1)/n}}$; $n \in \mathbb{N}$, take x^n common & put $1+x^{-n}=t^n$

(v) $\int \int \frac{dx}{x^n(1+x^n)^{1/n}}$; take x^n common as x and put $1+x^{-n}=t$.

Self Practice Problems :

(3) Evaluate : $\int \frac{\sec^2 x}{1+\tan x} dx$

(4) Evaluate : $\int \frac{\sin(\ln x)}{x} dx$

Ans. (3) $\ln |1 + \tan x| + C$

(4) $-\cos(\ln x) + C$

Integration by Parts : Product of two functions $f(x)$ and $g(x)$ can be integrated using formula :

$$\int (f(x) \cdot g(x)) dx = f(x) \int (g(x)) dx - \int \left(\frac{d}{dx}(f(x)) \cdot \int (g(x)) dx \right) dx$$

(i) when you find integral $\int g(x) dx$ then it will **not** contain arbitrary constant.

(ii) $\int g(x) dx$ should be taken as same at both places.

(iii) The choice of $f(x)$ and $g(x)$ can be decided by ILATE guideline. the function will come later is taken an integral function ($g(x)$).

I $\rightarrow$ Inverse function
L $\rightarrow$ Logarithmic function
A $\rightarrow$ Algebraic function
T $\rightarrow$ Trigonometric function
E $\rightarrow$ Exponential function

Example # 13 : Evaluate : $\int \sec^{-1} x dx$

Solution : Put $\sec^{-1} x = t$ so that $x = \sec t$ and $dx = \sec t \tan t dt$

$$\therefore \int \sec^{-1} x dx = \int t (\sec t \tan t) dt = t(\sec t) - \int 1 \cdot \sec t dt$$

$$= t \sec t - \ln |\sec t + \tan t| + C$$

$$= t \sec t - \ln |\sec t + \sqrt{\sec^2 t - 1}| + C = x(\sec^{-1} x) - \ln |x + \sqrt{x^2 - 1}| + C$$



Example # 14 : Evaluate : $\int x \ln(1+x) dx$

Solution : Let $I = \int x \ln(1+x) dx = \frac{x^2}{2} \cdot \ln(x+1) - \int \frac{1}{x+1} \cdot \frac{x^2}{2} dx$

$$= \frac{x^2}{2} \ln(x+1) - \frac{1}{2} \int \frac{x^2}{x+1} dx = \frac{x^2}{2} \ln(x+1) - \frac{1}{2} \int \frac{x^2-1+1}{x+1} dx$$

$$= \frac{x^2}{2} \ln(x+1) - \frac{1}{2} \int \left(\frac{x^2-1}{x+1} + \frac{1}{x+1} \right) dx = \frac{x^2}{2} \ln(x+1) - \frac{1}{2} \int \left((x-1) + \frac{1}{x+1} \right) dx$$

$$= \frac{x^2}{2} \ln(x+1) - \frac{1}{2} \left[\frac{x^2}{2} - x + \ln|x+1| \right] + C$$

Example # 15 : Evaluate : $\int e^{2x} \sin 3x dx$

Solution : Let $I = \int e^{2x} \sin 3x dx$

$$= e^{2x} \left(-\frac{\cos 3x}{3} \right) - \int 2e^{2x} \left(-\frac{\cos 3x}{3} \right) dx = -\frac{1}{3} e^{2x} \cos 3x + \frac{2}{3} \int e^{2x} \cos 3x dx$$

$$= -\frac{1}{3} e^{2x} \cos 3x + \frac{2}{3} \left[e^{2x} \frac{\sin 3x}{3} - \int 2e^{2x} \frac{\sin 3x}{3} dx \right]$$

$$= -\frac{1}{3} e^{2x} \cos 3x + \frac{2}{9} e^{2x} \sin 3x - \frac{4}{9} \int e^{2x} \sin 3x dx$$

$$\Rightarrow I = -\frac{1}{3} e^{2x} \cos 3x + \frac{2}{9} e^{2x} \sin 3x - \frac{4}{9} I \Rightarrow I + \frac{4}{9} I = \frac{e^{2x}}{9} (2 \sin 3x - 3 \cos 3x)$$

$$\Rightarrow \frac{13}{9} I = \frac{e^{2x}}{9} (2 \sin 3x - 3 \cos 3x) \Rightarrow I = \frac{e^{2x}}{13} (2 \sin 3x - 3 \cos 3x) + C$$

Note :

(i) $\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$ (ii) $\int [f(x) + x f'(x)] dx = x f(x) + C$

Example # 16 : Evaluate : $\int e^x \frac{(x^2 - 2x + 2)}{(x^2 + 2)^2} dx$

Solution : Given integral = $\int e^x \frac{(x^2 - 2x + 2)}{(x^2 + 2)^2} dx = \int e^x \left\{ \frac{1}{x^2 + 2} + \frac{(-2x)}{(x^2 + 2)^2} \right\} = \frac{e^x}{x^2 + 2} + C$

Example # 17 : Evaluate : $\int e^x \left(\frac{1 - \sin x}{1 - \cos x} \right) dx$

Solution : Given integral = $\int e^x \left(\frac{1 - 2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \sin^2 \frac{x}{2}} \right) dx$

$$= \int e^x \left(\frac{1}{2} \operatorname{cosec}^2 \frac{x}{2} - \cot \frac{x}{2} \right) dx = -e^x \cot \frac{x}{2} + C$$





Example # 18 : Evaluate : $\int \left[\ln(\ln x) + \frac{1}{(\ln x)^2} \right] dx$

Solution : Let $I = \int \left[\ln(\ln x) + \frac{1}{(\ln x)^2} \right] dx$ {put $x = e^t \Rightarrow dx = e^t dt$ }
 $\therefore I = \int e^t \left(\ln t + \frac{1}{t^2} \right) dt = \int e^t \left(\ln t - \frac{1}{t} + \frac{1}{t} + \frac{1}{t^2} \right) dt$
 $= e^t \left(\ln t - \frac{1}{t} \right) + C = x \left[\ln(\ln x) - \frac{1}{\ln x} \right] + C$

Self Practice Problems :

(5) Evaluate : $\int x \sin x \, dx$

(6) Evaluate : $\int x^2 e^x \, dx$

Ans. (5) $-x \cos x + \sin x + C$ (6) $x^2 e^x - 2x e^x + 2e^x + C$

Integration of type $\int \frac{dx}{ax^2 + bx + c}$, $\int \frac{dx}{\sqrt{ax^2 + bx + c}}$, $\int \sqrt{ax^2 + bx + c} \, dx$

Express $ax^2 + bx + c$ in the form of perfect square & then apply the standard results.

Example # 19 : Evaluate : $\int \sqrt{x^2 + 2x + 5} \, dx$

Solution : We have,
 $\int \sqrt{x^2 + 2x + 5} \, dx = \int \sqrt{x^2 + 2x + 1 + 4} \, dx = \int \sqrt{(x+1)^2 + 2^2} \, dx$
 $= \frac{1}{2} (x+1) \sqrt{(x+1)^2 + 2^2} + \frac{1}{2} \cdot (2)^2 \ln |(x+1) + \sqrt{(x+1)^2 + 2^2}| + C$
 $= \frac{1}{2} (x+1) \sqrt{x^2 + 2x + 5} + 2 \ln |(x+1) + \sqrt{x^2 + 2x + 5}| + C$

Example # 20 : Evaluate : $\int \frac{dx}{\sqrt{2-6x-9x^2}}$

Solution : $\int \frac{dx}{\sqrt{2-6x-9x^2}} \, dx = \int \frac{1}{\sqrt{3-(3x+1)^2}} \, dx = \frac{1}{3} \sin^{-1} \left(\frac{3x+1}{\sqrt{3}} \right) + C$

Self Practice Problems :

(7) Evaluate : $\int \frac{1}{2x^2 + x - 1} \, dx$

(8) Evaluate : $\int \frac{8x-11}{\sqrt{5+2x-x^2}} \, dx$

Ans. (7) $\frac{1}{3} \ln \left| \frac{2x-1}{2x+2} \right| + C$

(8) $-8 \sqrt{5+2x-x^2} - 3 \sin^{-1} \frac{x-1}{\sqrt{6}} + C$

Integration of type

$\int \frac{px + q}{ax^2 + bx + c} \, dx$, $\int \frac{px + q}{\sqrt{ax^2 + bx + c}} \, dx$, $\int (px + q) \sqrt{ax^2 + bx + c} \, dx$

Express $px + q = A$ (differential co-efficient of denominator) + B.



Example # 21 : Evaluate : $\int \frac{2x-3}{x^2+3x-18} dx$

Solution : Let $2x-3 = \lambda \frac{d}{dx} (x^2+3x-18) + \mu$

Then $2x-3 = \lambda (2x+3) + \mu$

Comparing the coefficients of like power of x , we get.

$2\lambda = 2$, and $3\lambda + \mu = -3 \Rightarrow \lambda = 1$ and $\mu = -6$

$$\begin{aligned} \text{So, } \int \frac{2x-3}{x^2+3x-18} dx &= \int \frac{2x+3-6}{x^2+3x-18} dx = \int \frac{2x+3}{x^2+3x-18} dx - 6 \int \frac{1}{x^2+3x-18} dx \\ &= \ell n|x^2+3x-18| - 6 \int \frac{1}{x^2+3x+\frac{9}{4}-\frac{9}{4}-18} dx = \ell n|x^2+3x-18| - 6 \int \frac{1}{\left(x+\frac{3}{2}\right)^2 - \left(\frac{9}{2}\right)^2} dx \end{aligned}$$

$$= \ell n|x^2+3x-18| - 6 \cdot \frac{1}{2\left(\frac{9}{2}\right)} \ell n \left| \frac{x+\frac{3}{2}-\frac{9}{2}}{x+\frac{3}{2}+\frac{9}{2}} \right| + C = \ell n|x^2+3x-18| - \frac{2}{3} \ell n \left| \frac{x-3}{x+6} \right| + C$$

Example # 22 : Evaluate : $\int \frac{2x+3}{\sqrt{x^2+4x+1}} dx$

Solution :

$$\begin{aligned} \int \frac{2x+3}{\sqrt{x^2+4x+1}} dx &= \int \frac{(2x+4)-1}{\sqrt{x^2+4x+1}} dx = \int \frac{2x+4}{\sqrt{x^2+4x+1}} dx - \int \frac{1}{\sqrt{x^2+4x+1}} dx \\ &= \int \frac{dt}{\sqrt{t}} - \int \frac{1}{\sqrt{(x+2)^2 - (\sqrt{3})^2}} dx, \quad \text{where } t = (x^2+4x+1) \text{ for 1st integral} \\ &= 2\sqrt{t} - \ell n|(x+2) + \sqrt{x^2+4x+1}| + C = 2\sqrt{x^2+4x+1} - \ell n|x+2+\sqrt{x^2+4x+1}| + C \end{aligned}$$

Example # 23 : Evaluate : $\int x\sqrt{1+x-x^2} dx$

Solution : Let $x = \lambda \cdot \frac{d}{dx} (1+x-x^2) + \mu$.

$\Rightarrow x = \lambda (1-2x) + \mu$

Comparing the coefficients of like powers of x , we get

$1 = -2\lambda$ and $\lambda + \mu = 0 \Rightarrow \lambda = -\frac{1}{2}$ and $\mu = \frac{1}{2} \therefore x = -\frac{1}{2} (1-2x) + \frac{1}{2}$

so, $\int x\sqrt{1+x-x^2} dx$

$$\begin{aligned} &= \int \left\{ -\frac{1}{2}(1-2x) + \frac{1}{2} \right\} \sqrt{1+x-x^2} dx = -\frac{1}{2} \int (1-2x)\sqrt{1+x-x^2} dx + \frac{1}{2} \int \sqrt{1+x-x^2} dx \\ &= -\frac{1}{2} \int \sqrt{1+x-x^2} d(1+x-x^2) + \frac{1}{2} \int \sqrt{\left(\frac{\sqrt{5}}{2}\right)^2 - \left(x-\frac{1}{2}\right)^2} dx, \\ &= -\frac{1}{3} (1+x-x^2)^{3/2} + \frac{1}{2} \left[\frac{1}{2} \left(x-\frac{1}{2}\right) \sqrt{\left(\frac{\sqrt{5}}{2}\right)^2 - \left(x-\frac{1}{2}\right)^2} + \frac{1}{2} \left(\frac{\sqrt{5}}{2}\right)^2 \sin^{-1} \frac{x-1/2}{\sqrt{5}/2} \right] + C \\ &= -\frac{1}{3} (1+x-x^2)^{3/2} + \frac{1}{2} \left[\left(x-\frac{1}{2}\right) \sqrt{1+x-x^2} + \frac{5}{8} \sin^{-1} \left(\frac{2x-1}{\sqrt{5}} \right) \right] + C \end{aligned}$$



**Self Practice Problems :**

(9) Evaluate : $\int \frac{3-4x}{2x^2-3x+1} dx$

(10) Evaluate : $\int \frac{6x-5}{\sqrt{3x^2-5x+1}} dx$

(11) Evaluate : $\int (x-1)\sqrt{1+x+x^2} dx$

Ans. (9) $-\ln|2x^2-3x+1| + C$

(10) $2\sqrt{3x^2-5x+1} + C$

(11) $\frac{1}{3}(x^2+x+1)^{3/2} - \frac{3}{8}(2x+1)\sqrt{1+x+x^2} - \frac{9}{16}\log(2x+1+2\sqrt{x^2+x+1}) + C$

Integration of Rational Algebraic Functions by using Partial Fractions:**PARTIAL FRACTIONS :**

If $f(x)$ and $g(x)$ are two polynomials, then $\frac{f(x)}{g(x)}$ defines a rational algebraic function of x .

If degree of $f(x) <$ degree of $g(x)$, then $\frac{f(x)}{g(x)}$ is called a proper rational function.

If degree of $f(x) \geq$ degree of $g(x)$ then $\frac{f(x)}{g(x)}$ is called an improper rational function.

If $\frac{f(x)}{g(x)}$ is an improper rational function, we divide $f(x)$ by $g(x)$ so that the rational function $\frac{f(x)}{g(x)}$ is expressed in the form $\phi(x) + \frac{\Psi(x)}{g(x)}$, where $\phi(x)$ and $\Psi(x)$ are polynomials such that the degree of $\Psi(x)$ is less than that of $g(x)$. Thus, $\frac{f(x)}{g(x)}$ is expressible as the sum of a polynomial and a proper rational function.

CASE-I $\frac{ax^2+bx+c}{(x-\alpha)(x-\beta)(x-\gamma)} = \frac{A}{x-\alpha} + \frac{B}{x-\beta} + \frac{C}{x-\gamma}$

CASE-II $\frac{ax^2+bx+c}{(x-\alpha)(x-\beta)^2} = \frac{A}{x-\alpha} + \frac{B}{x-\beta} + \frac{C}{(x-\beta)^2}$

CASE-III $\frac{ax^2+bx+c}{(x-\alpha)(x^2+\beta^2)} = \frac{A}{x-\alpha} + \frac{Bx+C}{x^2+\beta^2}$

where A, B, C can be evaluated by substitution or by comparing coefficients.

Example # 24 : Resolve $\frac{1}{2x^3+3x^2-3x-2}$ into partial fractions.

Solution : We have, $\frac{1}{2x^3+3x^2-3x-2} = \frac{1}{(x-1)(x+2)(2x+1)}$

Let $\frac{1}{2x^3+3x^2-3x-2} = \frac{A}{x-1} + \frac{B}{x+2} + \frac{C}{2x+1}$. Then,

$\Rightarrow 1 = A(x+2)(2x+1) + B(x-1)(2x+1) + C(x-1)(x+2) \quad \dots(i)$

Putting $x-1=0$ or $x=1$ in (i), we get $\Rightarrow A = \frac{1}{9}$,

Putting $x=-2$ in (i), we obtain $B = \frac{1}{9}$

Putting $x = -\frac{1}{2}$ in (i), we obtain $C = -\frac{4}{9}$

$\therefore \frac{1}{2x^3+3x^2-3x-2} = \frac{1}{(x-1)(x+2)(2x+1)} = \frac{1}{9(x-1)} + \frac{1}{9(x+2)} - \frac{4}{9(2x+1)}$





Example # 25 : Resolve $\frac{x^3 - 6x^2 + 10x - 2}{x^2 - 5x + 6}$ into partial fractions.

Solution : Here the given function is an improper rational function. On dividing we get

$$\frac{x^3 - 6x^2 + 10x - 2}{x^2 - 5x + 6} = x - 1 + \frac{(-x + 4)}{(x^2 - 5x + 6)} \quad \dots\dots\dots(i)$$

we have, $\frac{-x + 4}{x^2 - 5x + 6} = \frac{-x + 4}{(x - 2)(x - 3)}$

So, let $\frac{-x + 4}{(x - 2)(x - 3)} = \frac{A}{x - 2} + \frac{B}{x - 3}$, then

$$-x + 4 = A(x - 3) + B(x - 2) \quad \dots\dots\dots(ii)$$

Putting $x - 3 = 0$ or $x = 3$ in (ii), we get

$$1 = B(1) \Rightarrow B = 1.$$

Putting $x - 2 = 0$ or $x = 2$ in (ii), we get

$$2 = A(2 - 3) \Rightarrow A = -2$$

$$\therefore \frac{-x + 4}{(x - 2)(x - 3)} = \frac{-2}{x - 2} + \frac{1}{x - 3}$$

$$\text{Hence } \frac{x^3 - 6x^2 + 10x - 2}{x^2 - 5x + 6} = x - 1 - \frac{2}{x - 2} + \frac{1}{x - 3}$$

Example # 26 : Evaluate : $\int \frac{3x + 1}{(x - 1)^3(x + 1)} dx$

Solution : Let $\frac{3x + 1}{(x - 1)^3(x + 1)} = \frac{A}{x + 1} + \frac{B}{(x - 1)} + \frac{C}{(x - 1)^2} + \frac{D}{(x - 1)^3} \quad \dots\dots\dots(i)$

Multiplying both sides by $(x + 1)$ and then putting $x = -1$, we get

$$A = \frac{-2}{(-2)^3} = \frac{1}{4}$$

Multiplying both sides by $(x - 1)^3$ and then putting $x = 1$, we get

$$D = \frac{4}{2} = 2$$

From (i), we get

$$3x + 1 = A(x - 1)^3 + B(x - 1)^2(x + 1) + C(x - 1)(x + 1) + D(x + 1)$$

putting $x = 0$, we get

$$1 = -A + B - C + D$$

$$\Rightarrow 1 = -\frac{1}{4} + B - C + 2 \Rightarrow B - C = \frac{-3}{4}$$

Putting $x = 2$, we get

$$7 = A + 3B + 3C + 3D$$

$$\Rightarrow 7 = \frac{1}{4} + 3B + 3C + 6 \Rightarrow 3B + 3C = \frac{3}{4} \Rightarrow B + C = \frac{1}{4}$$

$$\text{Solving } B + C = \frac{1}{4} \text{ and } B - C = \frac{-3}{4}, \text{ we get } B = -\frac{1}{4}, C = \frac{1}{2}$$

Substituting the values of A, B, C and D in (i), we get

$$\Rightarrow \frac{3x + 1}{(x - 1)^3(x + 1)} = \frac{1}{4} \cdot \frac{1}{x + 1} - \frac{1}{4(x - 1)} + \frac{1}{2(x - 1)^2} + \frac{2}{(x - 1)^3}$$

$$\Rightarrow \int \frac{3x + 1}{(x - 1)^3(x + 1)} dx = \frac{1}{4} \int \frac{1}{x + 1} dx - \frac{1}{4} \int \frac{1}{x - 1} dx + \frac{1}{2} \int \frac{1}{(x - 1)^2} dx + 2 \int \frac{1}{(x - 1)^3} dx$$

$$= \frac{1}{4} \ln|x + 1| - \frac{1}{4} \ln|x - 1| - \frac{1}{2(x - 1)} - \frac{1}{(x - 1)^2} + C$$





Example # 27 : Evaluate : $\int \frac{1}{\sin x(2\cos^2 x - 1)} dx$

Solution : Putting $\cos x = t$, we get

$$I = \int \frac{1}{\sin x(2\cos^2 x - 1)} dx = \int \frac{1}{\sin x(2t^2 - 1)} \times -\frac{dt}{\sin x} = -\int \frac{1}{(1-t^2)(2t^2 - 1)} dt$$

$$\therefore I = -\int \left(\frac{1}{1-t^2} + \frac{2}{2t^2 - 1} \right) dt = -\int \frac{1}{1-t^2} dt - 2 \int \frac{1}{2t^2 - 1} dt$$

$$= -\frac{1}{2} \ln \left| \frac{1+t}{1-t} \right| - \frac{\sqrt{2}}{2} \ln \left| \frac{\sqrt{2}t-1}{\sqrt{2}t+1} \right| + C = -\frac{1}{2} \ln \left| \frac{1+\cos x}{1-\cos x} \right| - \frac{1}{\sqrt{2}} \ln \left| \frac{\sqrt{2}\cos x - 1}{\sqrt{2}\cos x + 1} \right| + C$$

Example # 28 : Resolve $\frac{2x-3}{(x-1)(x^2+1)^2}$ into partial fractions.

Solution : Let $\frac{2x-3}{(x-1)(x^2+1)^2} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{(x^2+1)^2}$. Then,

$$2x-3 = A(x^2+1)^2 + (Bx+C)(x-1)(x^2+1) + (Dx+E)(x-1) \quad \dots(i)$$

Putting $x = 1$ in (i), we get $-1 = A(1+1)^2 \Rightarrow A = -\frac{1}{4}$

Comparing coefficients of like powers of x on both side of (i), we have

$$A+B=0, C-B=0, 2A+B-C+D=0, C+E-B-D=2 \text{ and } A-C-E=-3.$$

Putting $A = -\frac{1}{4}$ and solving these equations, we get

$$B = \frac{1}{4}, C = \frac{1}{4} \text{ and } E = \frac{5}{2} \therefore \frac{2x-3}{(x-1)(x^2+1)^2} = \frac{-1}{4(x-1)} + \frac{x+1}{4(x^2+1)} + \frac{x+5}{2(x^2+1)^2}$$

Example # 29 : Resolve $\frac{2x}{x^3-1}$ into partial fractions.

Solution : We have, $\frac{2x}{x^3-1} = \frac{2x}{(x-1)(x^2+x+1)}$

So, let $\frac{2x}{(x-1)(x^2+x+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+x+1}$.

Then, $2x = A(x^2+x+1) + (Bx+C)(x-1) \dots(i)$

Putting $x-1 = 0$ or, $x = 1$ in (i), we get $2 = 3A \Rightarrow A = \frac{2}{3}$

Putting $x = 0$ in (i), we get $A - C = 0 \Rightarrow C = A = \frac{2}{3}$

Putting $x = -1$ in (i), we get $-2 = A + 2B - 2C \Rightarrow -2 = \frac{2}{3} + 2B - \frac{4}{3} \Rightarrow B = -\frac{2}{3}$

$$\therefore \frac{2x}{x^3-1} = \frac{2}{3} \cdot \frac{1}{x-1} + \frac{(-2/3)x + 2/3}{x^2+x+1} \text{ or } \frac{2x}{x^3-1} = \frac{2}{3} \cdot \frac{1}{x-1} + \frac{2}{3} \cdot \frac{1-x}{x^2+x+1}$$

Self Practice Problems :

(12) (i) Evaluate : $\int \frac{1}{(x+2)(x+3)} dx$ (ii) Evaluate : $\int \frac{dx}{(x+1)(x^2+1)}$

Ans. (12) (i) $\ln \left| \frac{x+2}{x+3} \right| + C$ (ii) $\frac{1}{2} \ln |x+1| - \ln(x^2+1) + \frac{1}{2} \tan^{-1}(x) + C$

**Integration of type**

$$\int \frac{x^2 \pm 1}{x^4 + Kx^2 + 1} dx \text{ where } K \text{ is any constant.}$$

Divide Nr & Dr by x^2 & put $x \mp \frac{1}{x} = t$.

Example # 30 : Evaluate $\int \frac{x^2 + 4}{x^4 + 16} dx$

Solution :
$$\int \frac{x^2 + 4}{x^4 + 16} dx = \int \frac{1 + \frac{4}{x^2}}{x^2 + \frac{16}{x^2}} dx = \int \frac{1}{\left(x - \frac{4}{x}\right)^2 + 8} d\left(x - \frac{4}{x}\right) = \int \frac{dt}{t^2 + (2\sqrt{2})^2},$$

where $t = x - \frac{4}{x} = \frac{1}{2\sqrt{2}} \tan^{-1} \left(\frac{t}{2\sqrt{2}} \right) + C = \frac{1}{2\sqrt{2}} \tan^{-1} \left(\frac{x^2 - 4}{2\sqrt{2}x} \right) + C$

Example # 31 : Evaluate : $\int \frac{x-1}{(x+1)\sqrt{x^3+x^2+x}} dx$

Solution :
$$\Rightarrow I = \int \frac{x^2 - 1}{(x+1)^2 \sqrt{x^3+x^2+x}} dx \quad \left[\begin{array}{l} \text{Multiplying the} \\ \text{Nr and Dr by } (x+1) \end{array} \right]$$

$$\Rightarrow I = \int \frac{(x^2 - 1)}{(x^2 + 2x + 1)\sqrt{x^3+x^2+x}} dx$$

$$\Rightarrow I = \int \frac{1 - \frac{1}{x^2}}{\left(x + \frac{1}{x} + 2\right)\sqrt{x + \frac{1}{x} + 1}} dx \quad \left[\text{Dividing Nr and Dr by } x^2 \right]$$

$$\Rightarrow I = \int \frac{2t \, dt}{(t^2 + 1)\sqrt{t^2}} \text{ where, } x + \frac{1}{x} + 1 = t^2 \Rightarrow I = 2 \int \frac{1}{t^2 + 1} dt \Rightarrow I = 2 \tan^{-1}(t) + C$$

$$\Rightarrow I = 2 \tan^{-1} \sqrt{x + \frac{1}{x} + 1} + C$$

Self Practice Problems :

(13) Evaluate : $\int \frac{x^2 - 1}{x^4 - 7x^2 + 1} dx$

(14) Evaluate : $\int \sqrt{\tan x} \, dx$

Ans. (13) $\frac{1}{6} \ln \left| \frac{x + \frac{1}{x} - 3}{x + \frac{1}{x} + 3} \right| + C$

(14) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{y}{\sqrt{2}} \right) + \frac{1}{2\sqrt{2}} \ln \left| \frac{y - \sqrt{2}}{y + \sqrt{2}} \right| + C$

where $y = \sqrt{\tan x} - \frac{1}{\sqrt{\tan x}}$

Integration of type

$$\int \frac{dx}{(ax + b)\sqrt{px + q}} \text{ OR } \int \frac{dx}{(ax^2 + bx + c)\sqrt{px + q}}.$$

Put $px + q = t^2$.



Example # 32 : Evaluate : $\int \frac{dx}{(x-4)\sqrt{x+5}}$

Solution : Let $I = \int \frac{dx}{(x-4)\sqrt{x+5}}$ {Put $x+5 = t^2 \Rightarrow dx = 2t dt$ }

$$\therefore I = \int \frac{2t dt}{(t^2-9)} = \frac{2}{6} \ln \left| \frac{t-3}{t+3} \right| + C = \frac{1}{3} \ln \left| \frac{\sqrt{x+5}-3}{\sqrt{x+5}+3} \right| + C$$

Example # 33 : Evaluate : $\int \frac{dx}{(x^2+3x+2)\sqrt{x+4}}$

Solution : Let $I = \int \frac{dx}{(x^2+3x+2)\sqrt{x+4}}$

Putting $x+4 = t^2$, and $dx = 2t dt$, we get $I = \int \frac{2t dt}{\{(t^2-4)^2+3(t^2-4)+2\}\sqrt{t^2}}$

$$\Rightarrow 2 \int \frac{dt}{t^4-5t^2+6} = 2 \int \frac{dt}{(t^2-2)(t^2-3)} = 2 \int \left[\frac{1}{t^2-3} - \frac{1}{t^2-2} \right] dt$$

$$= \frac{1}{\sqrt{3}} \ln \left| \frac{t-\sqrt{3}}{t+\sqrt{3}} \right| - \frac{1}{\sqrt{2}} \ln \left| \frac{t-\sqrt{2}}{t+\sqrt{2}} \right| + C \text{ where } t = \sqrt{x+4}$$

Integration of type

$$\int \frac{dx}{(ax+b)px\sqrt{2qx+r}}, \text{ put } ax+b = \frac{1}{t}; \quad \int \frac{dx}{(ax^2+b)px\sqrt{2qx+r}}, \text{ put } x = \frac{1}{t}$$

Example # 34 : Evaluate $\int \frac{dx}{(x-1)x\sqrt{2x-1}}$

Solution : Let $I = \int \frac{dx}{(x-1)x\sqrt{2x-1}}$ {put $x-1 = \frac{1}{t} \Rightarrow dx = -\frac{1}{t^2} dt$ }

$$\Rightarrow I = \int \frac{-\frac{1}{t^2} dt}{\frac{1}{t} \sqrt{\left(\frac{1}{t}+1\right)^2 - \left(\frac{1}{t}+1\right)-1}} = \int \frac{dt}{\sqrt{-t^2+t+1}} = \int \frac{dt}{\sqrt{\left(\frac{\sqrt{5}}{2}\right)^2 - \left(t-\frac{1}{2}\right)^2}}$$

$$= -\sin^{-1} \left(\frac{t-\frac{1}{2}}{\frac{\sqrt{5}}{2}} \right) + C = -\sin^{-1} \left(\frac{2t-1}{\sqrt{5}} \right) + C, \text{ where } t = \frac{1}{x-1}$$

Example # 35 : Evaluate $\int \frac{dx}{(1+x^2)\sqrt{1-x^2}}$

Solution : Put $x = \frac{1}{t} \Rightarrow dx = -\frac{1}{t^2} dt \Rightarrow I = - \int \frac{tdt}{(t^2+1)\sqrt{t^2-1}}$ {put $t^2-1 = y^2 \Rightarrow tdt = ydy$ }

$$\Rightarrow I = - \int \frac{y dy}{(y^2+2)y} = - \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{y}{\sqrt{2}} \right) + C$$

$$= - \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{1-x^2}}{\sqrt{2}x} \right) + C$$





Self Practice Problems :

- (15) Evaluate : $\int \frac{dx}{(x+2)\sqrt{x+1}}$ (16) Evaluate : $\int \frac{dx}{(x^2+5x+6)\sqrt{x+1}}$
- (17) Evaluate : $\int \frac{dx}{(x+1)\sqrt{1+x-x^2}}$ (18) Evaluate : $\int \frac{dx}{(2x^2+1)\sqrt{1-x^2}}$
- (19) Evaluate : $\int \frac{dx}{(x^2+2x+2)\sqrt{x^2+2x-4}}$

Ans. (15) $2 \tan^{-1}(\sqrt{x+1}) + C$ (16) $2 \tan^{-1}(\sqrt{x+1}) - \sqrt{2} \tan^{-1}\left(\frac{\sqrt{x+1}}{\sqrt{2}}\right) + C$

(17) $\sin^{-1}\left(\frac{\frac{3}{2} - \frac{1}{x+1}}{\frac{\sqrt{5}}{2}}\right) + C$ (18) $-\frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{\sqrt{1-x^2}}{\sqrt{3}x}\right) + C$

(19) $-\frac{1}{2\sqrt{6}} \ln\left(\frac{\sqrt{x^2+2x-4} - \sqrt{6}(x+1)}{\sqrt{x^2+2x-4} + \sqrt{6}(x+1)}\right) + C$

Integration of type

$\int \sqrt{\frac{x-\alpha}{\beta-x}} dx$ or $\int \sqrt{(x-\alpha)(\beta-x)} dx$; put $x = \alpha \cos^2 \theta + \beta \sin^2 \theta$

$\int \sqrt{\frac{x-\alpha}{x-\beta}} dx$ or $\int \sqrt{(x-\alpha)(x-\beta)} dx$; put $x = \alpha \sec^2 \theta - \beta \tan^2 \theta$

$\int \frac{dx}{\sqrt{(x-\alpha)(x-\beta)}}$; put $x-\alpha = t^2$ or $x-\beta = t^2$.

Self Practice Problems

- (20) Evaluate : $\int \sqrt{\frac{x-3}{x-4}} dx$ (21) Evaluate : $\int \frac{dx}{[(x-1)(2-x)]^{3/2}}$
- (22) Evaluate : $\int \frac{dx}{[(x+2)^8(x-1)^6]^{1/7}}$

Ans. (20) $\sqrt{(x-3)(x-4)} + \ln(\sqrt{x-3} + \sqrt{x-4}) + C$ (21) $2\left(\sqrt{\frac{x-1}{2-x}} - \sqrt{\frac{2-x}{x-1}}\right) + C$

(22) $\frac{7}{3} \left(\frac{x-1}{x+2}\right)^{1/7} + C$

Integration of trigonometric functions

(i) $\int \frac{dx}{a + b \sin^2 x}$ OR $\int \frac{dx}{a + b \cos^2 x}$ OR $\int \frac{dx}{a \sin^2 x + b \sin x \cos x + c \cos^2 x}$
Multiply Nr & Dr by $\sec^2 x$ & put $\tan x = t$.

(ii) $\int \frac{dx}{a + b \sin x}$ OR $\int \frac{dx}{a + b \cos x}$ OR $\int \frac{dx}{a + b \sin x + c \cos x}$

Convert sines & cosines into their respective tangents of half the angles and then, put $\tan \frac{x}{2} = t$

(iii) $\int \frac{a \cos x + b \sin x + c}{\ell \cos x + m \sin x + n} dx$.

Express Nr $\equiv A(\text{Dr}) + B(\text{Dr}) + C$ & proceed.





Example # 36 : Evaluate: $\int \frac{1 + \sin x}{\sin x(1 + \cos x)} dx$

Solution : Let $I = \int \frac{1 + \sin x}{\sin x(1 + \cos x)} dx$

$$\text{Putting } \sin x = \frac{2 \tan x/2}{1 + \tan^2 x/2} \text{ and } \cos x = \frac{1 - \tan^2 x/2}{1 + \tan^2 x/2},$$

we get

$$\begin{aligned} I &= \int \frac{1 + \frac{2 \tan x/2}{1 + \tan^2 x/2}}{\left(\frac{2 \tan x/2}{1 + \tan^2 x/2}\right) \left(1 + \frac{1 - \tan^2 x/2}{1 + \tan^2 x/2}\right)} dx = \int \frac{(1 + \tan^2 x/2 + 2 \tan x/2)(1 + \tan^2 x/2)}{2 \tan x/2 (1 + \tan^2 x/2 + 1 - \tan^2 x/2)} dx \\ &= \int \frac{(1 + \tan^2 x/2)^2 \sec^2 x/2}{4 \tan x/2} dx = \int \frac{1 + t^2 + 2t}{2t} dt, \text{ where } t = \tan \frac{x}{2} \\ &= \frac{1}{2} \int \left(\frac{1}{t} + t + 2\right) dt = \frac{1}{2} \left[\ln |t| + \frac{t^2}{2} + 2t \right] + C = \frac{1}{2} \left[\ln \left| \tan \frac{x}{2} \right| + \frac{\tan^2 x/2}{2} + 2 \tan x/2 \right] + C \end{aligned}$$

Example # 37 : Evaluate : $\int \frac{dx}{\sin x + \sqrt{3} \cos x}$

Solution : Let $1 = r \cos \theta$ and $\sqrt{3} = r \sin \theta \Rightarrow r = \sqrt{(1)^2 + (\sqrt{3})^2} = 2$

$$\tan \theta = \sqrt{3} \Rightarrow \theta = \pi/3$$

$$\begin{aligned} \therefore \int \frac{dx}{\sin x + \sqrt{3} \cos x} &= \frac{1}{r} \int \frac{dx}{\sin x \cos \theta + \cos x \sin \theta} = \frac{1}{r} \int \frac{dx}{\sin(x + \theta)} \\ &= \frac{1}{r} \int \operatorname{cosec}(x + \theta) dx = \frac{1}{r} \ln \left| \tan \left(\frac{x}{2} + \frac{\theta}{2} \right) \right| + C = \frac{1}{2} \ln \left| \tan \left(\frac{x}{2} + \frac{\pi}{6} \right) \right| + C \end{aligned}$$

Example # 38 : Evaluate : $\int \frac{3 \cos x + 2}{\sin x + 2 \cos x + 3} dx$

Solution : We have,

$$I = \int \frac{3 \cos x + 2}{\sin x + 2 \cos x + 3} dx$$

$$\text{Let } 3 \cos x + 2 = \lambda (\sin x + 2 \cos x + 3) + \mu (\cos x - 2 \sin x) + v$$

Comparing the coefficients of $\sin x$, $\cos x$ and constant term on both sides, we get

$$\lambda - 2\mu = 0, 2\lambda + \mu = 3, 3\lambda + v = 2 \quad \Rightarrow \quad \lambda = \frac{6}{5}, \mu = \frac{3}{5} \text{ and } v = -\frac{8}{5}$$

$$\therefore I = \int \frac{\lambda(\sin x + 2 \cos x + 3) + \mu(\cos x - 2 \sin x) + v}{\sin x + 2 \cos x + 3} dx$$

$$\Rightarrow I = \lambda \int dx + \mu \int \frac{\cos x - 2 \sin x}{\sin x + 2 \cos x + 3} dx + v \int \frac{1}{\sin x + 2 \cos x + 3} dx$$

$$\Rightarrow I = \lambda x + \mu \log |\sin x + 2 \cos x + 3| + v I_1$$

$$\text{where } I_1 = \int \frac{1}{\sin x + 2 \cos x + 3} dx$$





Putting, $\sin x = \frac{2 \tan x/2}{1 + \tan^2 x/2}$, $\cos x = \frac{1 - \tan^2 x/2}{1 + \tan^2 x/2}$, we get

$$I_1 = \int \frac{1}{\frac{2 \tan x/2}{1 + \tan^2 x/2} + \frac{2(1 - \tan^2 x/2)}{1 + \tan^2 x/2} + 3} dx = \int \frac{1 + \tan^2 x/2}{2 \tan x/2 + 2 - 2 \tan^2 x/2 + 3(1 + \tan^2 x/2)} dx$$

$$= \int \frac{\sec^2 x/2}{\tan^2 x/2 + 2 \tan x/2 + 5} dx$$

Putting $\tan \frac{x}{2} = t$ and $\frac{1}{2} \sec^2 \frac{x}{2} = dt$ or $\sec^2 \frac{x}{2} dx = 2 dt$, we get

$$I_1 = \int \frac{2dt}{t^2 + 2t + 5} = 2 \int \frac{dt}{(t+1)^2 + 2^2} = \frac{2}{2} \tan^{-1} \left(\frac{t+1}{2} \right) = \tan^{-1} \left(\frac{\tan \frac{x}{2} + 1}{2} \right)$$

$$\text{Hence, } I = \lambda x + \mu \log |\sin x + 2 \cos x + 3| + \nu \tan^{-1} \left(\frac{\tan \frac{x}{2} + 1}{2} \right) + C$$

$$\text{where } \lambda = \frac{6}{5}, \mu = \frac{3}{5} \text{ and } \nu = -\frac{8}{5}$$

Example # 39 : Evaluate : $\int \frac{dx}{1 + 3 \cos^2 x}$

Solution : Multiply Nr. & Dr. of given integral by $\sec^2 x$, we get

$$I = \int \frac{\sec^2 x}{\tan^2 x + 4} dx = \frac{1}{2} \tan^{-1} \left(\frac{\tan x}{2} \right) + C$$

Self Practice Problems :

(23) Evaluate : $\int \frac{4 \sin x + 5 \cos x}{5 \sin x + 4 \cos x} dx$

Ans. (23) $\frac{40}{41} x + \frac{9}{41} \log |5 \sin x + 4 \cos x| + C$

Integration of type $\int \sin^m x \cdot \cos^n x \, dx$

Case - I

If m and n are even natural number then converts higher power into higher angles.

Case - II

If at least one of m or n is odd natural number then if m is odd put $\cos x = t$ and vice-versa.

Case - III

When $m + n$ is a negative even integer then put $\tan x = t$.

Example # 40 : Evaluate : $\int \cos^5 x \sin^4 x \, dx$

Solution : Let $I = \int \cos^5 x \sin^4 x \, dx$ put $\sin x = t \Rightarrow \cos x \, dx = dt$

$$\Rightarrow I = \int (1 - t^2)^2 \cdot t^4 \cdot dt = \int (t^4 - 2t^2 + 1) t^4 dt = \int (t^8 - 2t^6 + t^4) dt$$

$$= \frac{t^9}{9} - \frac{2t^7}{7} + \frac{t^5}{5} + C, \text{ where } t = \sin x$$



Example # 41 : Evaluate : $\int \sec^{4/3} x \operatorname{cosec}^{8/3} x dx$

Solution : We have ,

$$I = \int \sec^{4/3} x \operatorname{cosec}^{8/3} x dx = \int \frac{1}{\cos^{4/3} x \sin^{8/3} x} dx = \int \cos^{-4/3} x \sin^{-8/3} x dx$$

divide Nr and Dr by $\cos^4 x$

$$= \int \frac{\sec^4 x}{\tan^{8/3} x} dx = \int \frac{(1 + \tan^2 x)}{\tan^{8/3} x} \sec^2 x dx = \int \frac{1 + \tan^2 x}{\tan^{8/3} x} d(\tan x) = \int \frac{1 + t^2}{t^{8/3}} dt \quad \text{where } t = \tan x$$

$$= \int (t^{-8/3} + t^{-2/3}) dt = \frac{-3}{5} t^{-5/3} + 3t^{1/3} + C = \frac{-3}{5} \tan^{-5/3} x + 3 \tan^{1/3} x + C$$

Example # 42 : Evaluate : $\int \sin^4 x \cos^2 x dx$

Solution :

$$\begin{aligned} \int \sin^4 x \cos^2 x dx &= \frac{1}{8} \int 4 \sin^2 x \cos^2 x \cdot 2 \sin^2 x dx = \frac{1}{8} \int \sin^2 2x (1 - \cos 2x) dx \\ &= \frac{1}{8} \int \sin^2 2x dx - \frac{1}{8} \int \sin^2 2x \cos 2x dx = \frac{1}{16} \int (1 - \cos 4x) dx - \frac{1}{48} (\sin 2x)^3 \\ &= \frac{x}{16} - \frac{\sin 4x}{64} - \frac{1}{48} (\sin 2x)^3 + C \end{aligned}$$

Reduction formula of $\int \tan^n x dx$, $\int \cot^n x dx$, $\int \sec^n x dx$, $\int \operatorname{cosec}^n x dx$

1. $I_n = \int \tan^n x dx = \int \tan^2 x \tan^{n-2} x dx = \int (\sec^2 x - 1) \tan^{n-2} x dx$

$$\Rightarrow I_n = \int \sec^2 x \tan^{n-2} x dx - I_{n-2} \Rightarrow I_n = \frac{\tan^{n-1} x}{n-1} - I_{n-2}, n \geq 2$$

2. $I_n = \int \cot^n x dx = \int \cot^2 x \cdot \cot^{n-2} x dx = \int (\operatorname{cosec}^2 x - 1) \cot^{n-2} x dx$

$$\Rightarrow I_n = \int \operatorname{cosec}^2 x \cot^{n-2} x dx - I_{n-2} \Rightarrow I_n = -\frac{\cot^{n-1} x}{n-1} - I_{n-2}, n \geq 2$$

3. $I_n = \int \sec^n x dx = \int \sec^2 x \sec^{n-2} x dx$

$$\Rightarrow I_n = \tan x \sec^{n-2} x - \int (\tan x)(n-2) \sec^{n-3} x \cdot \sec x \tan x dx.$$

$$\Rightarrow I_n = \tan x \sec^{n-2} x - (n-2) \int (\sec^2 x - 1) \sec^{n-2} x dx$$

$$\Rightarrow (n-1) I_n = \tan x \sec^{n-2} x + (n-2) I_{n-2} \Rightarrow I_n = \frac{\tan x \sec^{n-2} x}{n-1} + \frac{n-2}{n-1} I_{n-2}$$

4. $I_n = \int \operatorname{cosec}^n x dx = \int \operatorname{cosec}^2 x \operatorname{cosec}^{n-2} x dx$

$$\Rightarrow I_n = -\cot x \operatorname{cosec}^{n-2} x + \int (\cot x)(n-2) (-\operatorname{cosec}^{n-3} x \operatorname{cosec} x \cot x) dx$$

$$\Rightarrow -\cot x \operatorname{cosec}^{n-2} x - (n-2) \int \cot^2 x \operatorname{cosec}^{n-2} x dx$$

$$\Rightarrow I_n = -\cot x \operatorname{cosec}^{n-2} x - (n-2) \int (\operatorname{cosec}^2 x - 1) \operatorname{cosec}^{n-2} x dx$$

$$\Rightarrow (n-1) I_n = -\cot x \operatorname{cosec}^{n-2} x + (n-2) I_{n-2}$$

$$\Rightarrow I_n = \frac{\cot x \operatorname{cosec}^{n-2} x}{-(n-1)} + \frac{n-2}{n-1} I_{n-2}$$



Example # 43 : Obtain the reduction formula for $\int \cos^n x dx$

Solution : Let $I_n = \int \cos^n x dx$

$$I_n = \int \cos x (\cos x)^{n-1} dx$$

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$$I_n = (\sin x)(\cos x)^{n-1} - \int (n-1)(\cos x)^{n-2}(-\sin x) \sin x dx$$

$$I_n = (\sin x)(\cos x)^{n-1} + (n-1) \int (\cos x)^{n-2} (1 - \cos^2 x) dx$$

$$I_n = (\sin x)(\cos x)^{n-1} + (n-1) \int (\cos x)^{n-2} dx - (n-1) \int (\cos x)^n dx$$

$$I_n = (\sin x)(\cos x)^{n-1} + (n-1) I_{n-2} - (n-1) I_n$$

$$I_n + (n-1)I_n = (\sin x)(\cos x)^{n-1} + (n-1)I_{n-2}$$

$$I_n = \frac{(\sin x)(\cos x)^{n-1}}{n} + \frac{(n-1)}{n} I_{n-2}, n \geq 2$$

Self Practice Problems :

(24) Deduce the reduction formula for $I_n = \int \frac{dx}{(1+x^4)^n}$ and Hence evaluate $I_2 = \int \frac{dx}{(1+x^4)^2}$.

(25) If $I_{m,n} = \int (\sin x)^m (\cos x)^n dx$ then prove that

$$I_{m,n} = \frac{(\sin x)^{m+1} (\cos x)^{n-1}}{m+n} + \frac{n-1}{m+n} \cdot I_{m,n-2}$$

Ans.

$$(24) \quad I_n = \frac{x}{4(n-1)(1+x^4)^{n-1}} + \frac{4n-5}{4(n-1)} I_{n-1}$$

$$I_2 = \frac{x}{4(1+x^4)} + \left(\frac{1}{2\sqrt{2}} \tan^{-1} \left(\frac{x - \frac{1}{x}}{\sqrt{2}} \right) - \frac{1}{4\sqrt{2}} \ln \left(\frac{x + \frac{1}{x} - \sqrt{2}}{x + \frac{1}{x} + \sqrt{2}} \right) \right) + C$$

